Wazuh Setup and Configuration

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1 What is SIEM

SIEM (Security Information and Event Management) is a cybersecurity solution that collects, stores, and analyzes security logs from multiple devices such as servers, workstations, network equipment, and applications. By centralizing log data, SIEM enables security analysts to monitor systems, detect suspicious activities, investigate incidents, and respond to potential threats in real time.

Within a **Security Operations Center (SOC),** SIEM plays an important role by providing:

* Real-time monitoring of endpoints and network activities.
* Centralized collection of security logs.
* Threat detection and alert generation.
* Event correlation and incident investigation.

Among the many SIEM solutions available today, this project uses **Wazuh**, an open-source SIEM platform that provides enterprise-level security monitoring while remaining free and highly customizable.

2 Overview of Wazuh SIEM

Wazuh is an open-source Security Information and Event Management (SIEM) platform designed to monitor, detect, and respond to security threats. It combines endpoint security, log analysis, file integrity monitoring, vulnerability detection, compliance monitoring, and incident response into a single platform.

Some of the major capabilities of Wazuh include:

* Threat Detection
* Log Collection and Analysis
* File Integrity Monitoring (FIM)
* Security Configuration Assessment (SCA)
* Malware Detection
* Vulnerability Detection
* Incident Response
* Compliance Monitoring

Unlike many commercial SIEM platforms, Wazuh is completely open source, making it an excellent choice for learning SOC operations and building cybersecurity labs.

3 Wazuh Architecture

The Wazuh architecture consists of four major components that work together to monitor and analyze security events.

Wazuh Agent

The Wazuh Agent is installed on monitored endpoints such as Windows, Linux, or macOS systems. It continuously collects security logs, monitors file changes, and sends event data to the Wazuh Manager.

Wazuh Manager

The Wazuh Manager receives logs from all connected agents, analyzes the collected events using built-in detection rules, and generates security alerts whenever suspicious activity is identified.

Wazuh Indexer

The Wazuh Indexer stores all security events received from the Wazuh Manager. It indexes the data, allowing fast searches, filtering, and efficient retrieval of security information.

Wazuh Dashboard

The Wazuh Dashboard provides a web-based interface for visualizing alerts, monitoring endpoints, performing threat hunting, and investigating security incidents.

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4 Wazuh Installation

Unlike the reference implementation that uses the pre-built Wazuh virtual machine (OVA), this project installs Wazuh manually on an Ubuntu Server virtual machine using the official Wazuh installation documentation.

5 Configuration

Host Platform: UTM (Apple Silicon)
Host Operating System: macOS
Server Operating System: Ubuntu Server
Target Agent Operating System: Windows 11
Attacker Machine: Kali Linux

Step 1: Create the Ubuntu Server Virtual Machine

1. Create a new Ubuntu Server virtual machine using UTM.
2. Allocate the required CPU cores, memory, and storage.
3. Install Ubuntu Server.
4. Complete the initial Ubuntu setup.
5. Log in to the Ubuntu server.

Step 2: Update Ubuntu

Before installing Wazuh, update the Ubuntu packages using the following commands:

```
sudo apt update
sudo apt upgrade -y
```

This ensures that the operating system is fully updated before installing the SIEM platform.

Step 3: Install Wazuh

Instead of downloading a pre-configured virtual machine, the installation was performed using the official Wazuh documentation.

The following procedure was followed:

1. Open the official Wazuh installation documentation.
2. Navigate to the installation guide
3. Copy the official all-in-one installation command provided by Wazuh.
4. Paste the command into the Ubuntu terminal.
5. Execute the command.
6. Wait for the installation process to complete.

```
curl -sO https://packages.wazuh.com/4.14/wazuh-install.sh && sudo bash ./wazuh-install.sh -a
```

The installation script automatically installed:

* Wazuh Manager
* Wazuh Indexer
* Wazuh Dashboard

```
INFO: --- Summary ---
INFO: You can access the web interface https://<WAZUH_DASHBOARD_IP_ADDRESS>
    User: admin
    Password: <ADMIN_PASSWORD>
INFO: Installation finished.
```

default username and password ⬆️

After the installation completed successfully, all required Wazuh components were available on the Ubuntu server.

Step 4: Verify Wazuh Services

After the installation completed, the status of each Wazuh service was verified.

The following services were checked:

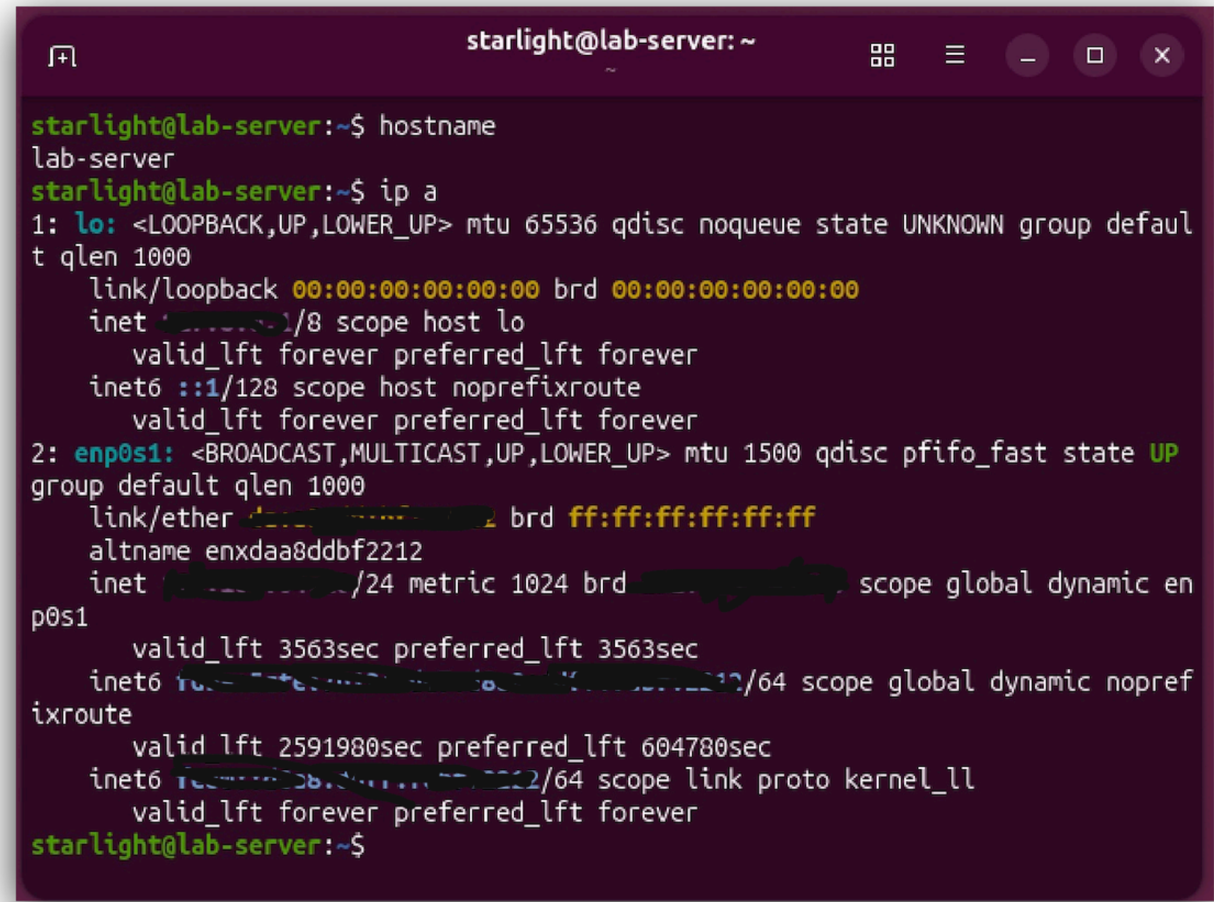
* Wazuh Manager
* Wazuh Dashboard
* Wazuh Indexer

If any service was not running, it was started using the appropriate systemctl command until all services became active.

Step 5: Obtain the Server IP Address

After confirming that all services were running, the Ubuntu server IP address was obtained using:

```
ip a
```



The IPv4 address assigned to the Ubuntu virtual machine was noted for accessing the dashboard and connecting Windows agents.

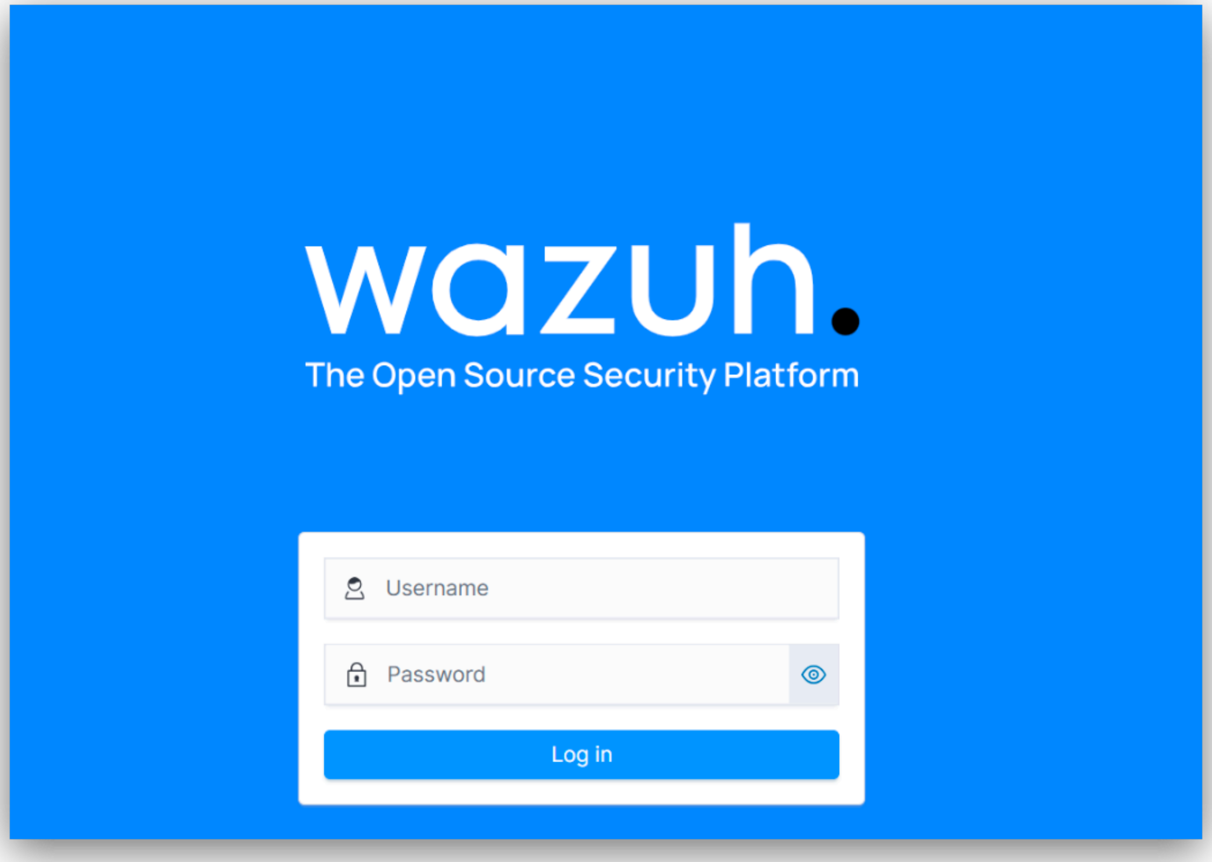
Step 6: Access the Wazuh Dashboard

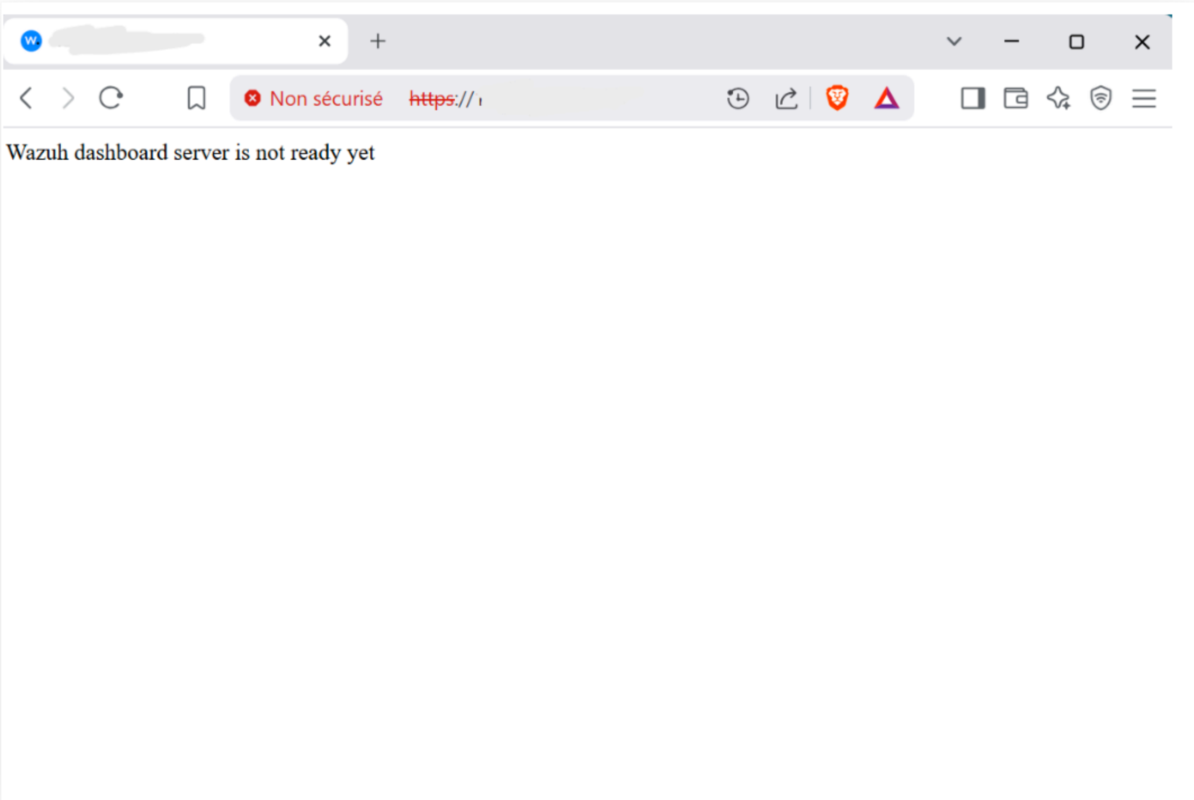
Using the Ubuntu server IP address, the Wazuh Dashboard was accessed through a web browser.

```
https://<IP_ADDRESS>
```

The Wazuh login page should open.

Wazuh Dashboard Interface





Wazuh Dashboard error: Server not ready

6 Wazuh Agent Installation and Configuration

6.1 What is the Wazuh Agent?

The Wazuh Agent is a lightweight application installed on monitored endpoints. It continuously collects system logs and security events before securely forwarding them to the Wazuh Manager for analysis.

6.2 Supported Platforms

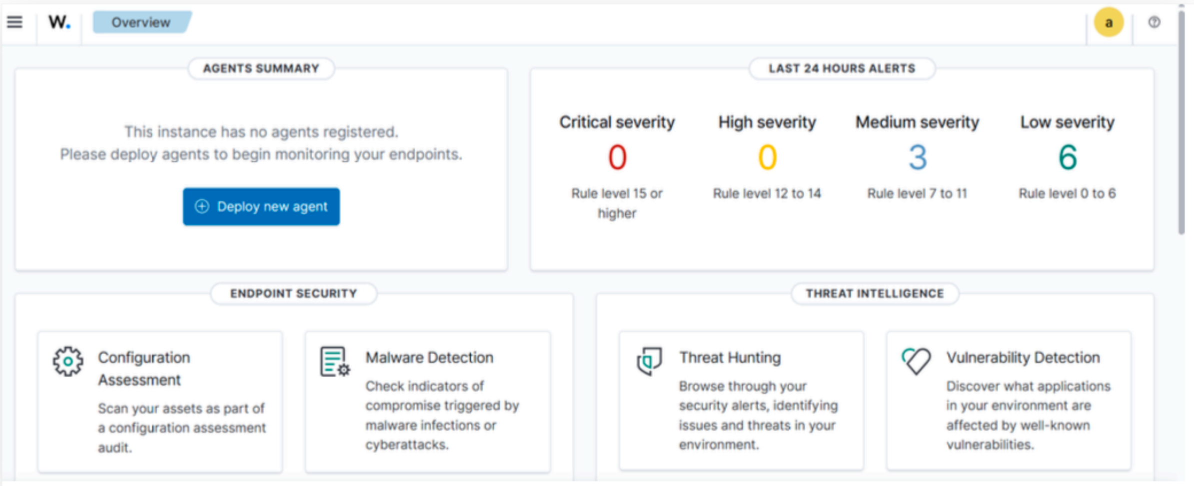
The Wazuh Agent supports multiple operating systems including:

* Windows
* Linux
* macOS

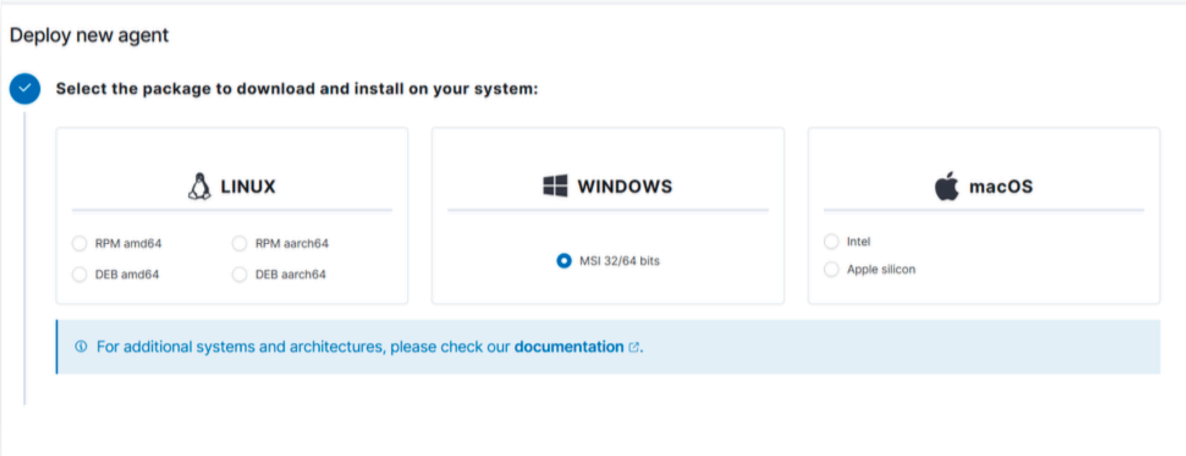
6.3 Deploying the Windows Agent

Step 1: Deploying

1. Open the Wazuh Dashboard.



2.Navigate to the Endpoints tab.
3.Click on Deploy new agent.



Step 2: Configure the Agent

The following information was entered:

• Enter the Server Address — the IP address of your Wazuh Manager.

• Specify a unique Agent Name (e.g., soc-user).
After entering the required information, Wazuh generated the installation commands for Windows.

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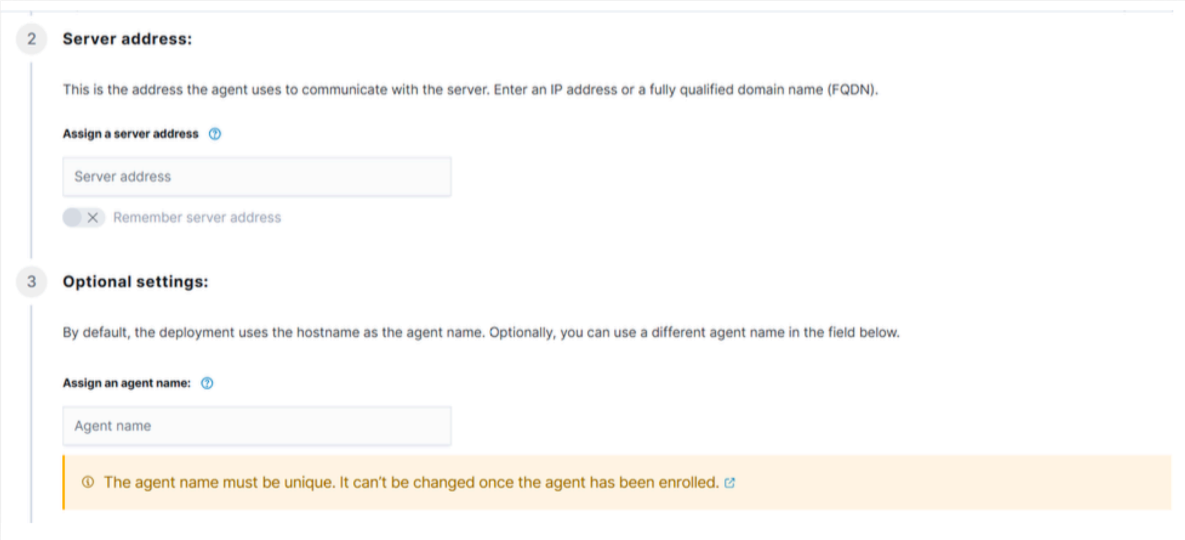

Step

Step 3: Register the Agent via PowerShell

After configuration, Wazuh will generate registration commands. These commands must be executed in PowerShell to link the agent with the manager.

- Open Windows PowerShell as Administrator.
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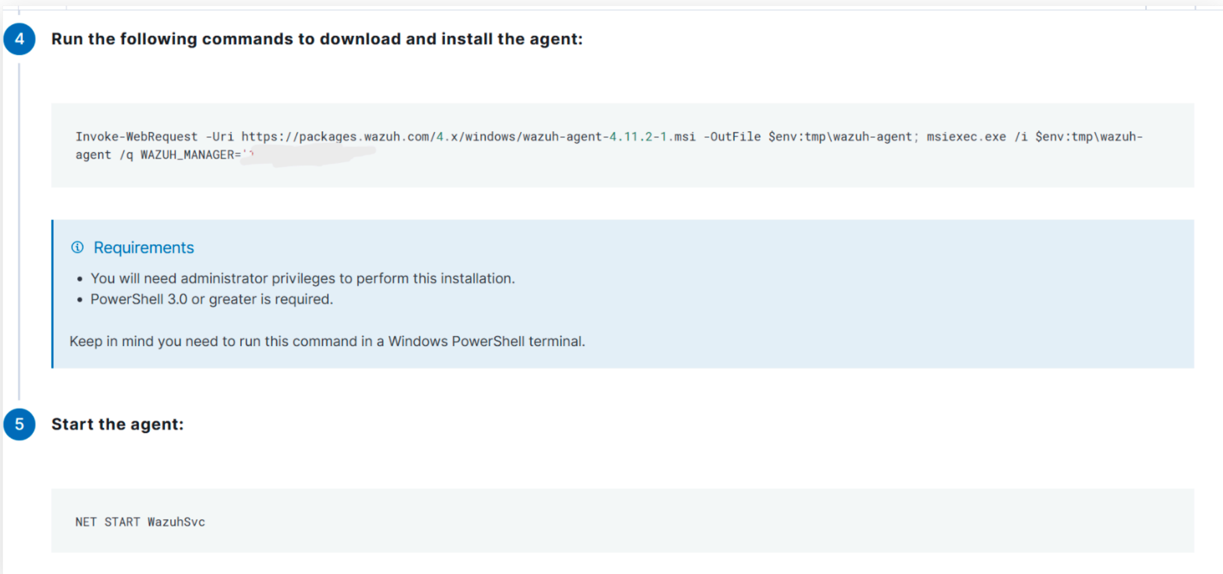
- Paste and execute the provided commands.

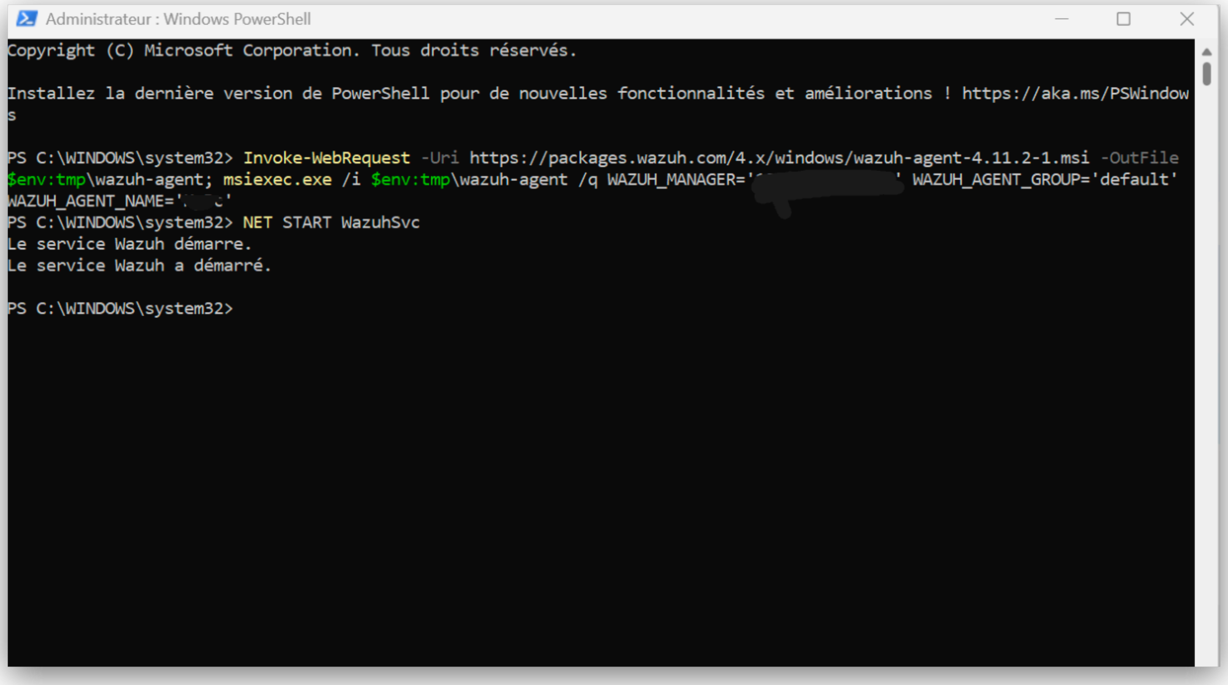


Step 4: Agent registration

On the Windows virtual machine:

1. Open Windows PowerShell as Administrator.
2. Copy the generated registration command from the Wazuh Dashboard.
3. Paste and execute the command.
4. Allow the installation to complete.
5. Start the Wazuh Agent service.

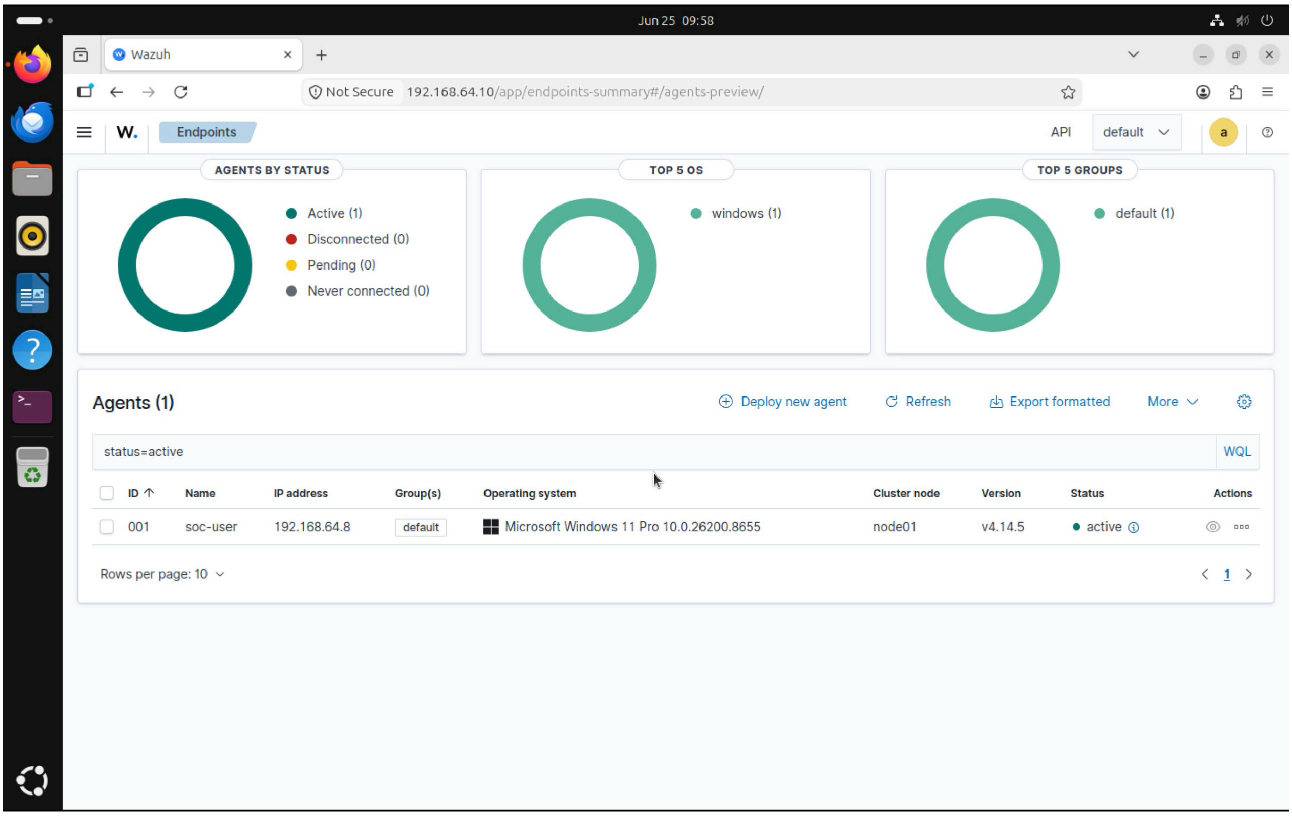
The Windows endpoint was successfully registered with the Wazuh Manage



After the registration process completed:

1. Return to the Wazuh Dashboard.
2. Navigate to **Endpoints**.
3. Verify that the Windows endpoint appears in the Agents list.
4. Confirm that the agent status changes to **Active**.

Once the status became Active, the Windows endpoint began forwarding security events to the Wazuh Manager successfully.



• Once installed, the agent uses a GUI for configuration, opening the log file, and starting or stopping the service.

• By default, all agent files are stored in C:\Program Files (x86)\ossec-agent after the installation.

7 Conclusion

This document described the installation and configuration of a Wazuh SIEM environment using Ubuntu Server and UTM virtualization. Unlike the pre-built virtual machine deployment, Wazuh was installed manually by following the official installation documentation and executing the provided installation script on Ubuntu Server.

After successfully installing the Wazuh Manager, Indexer, and Dashboard, the Windows 11 endpoint was deployed and registered using the Wazuh Agent. The successful connection between the server and endpoint established the foundation for future security monitoring, threat detection, and SOC laboratory exercises, which will be covered in subsequent documents.